

Problem Set 4 — STAT 380

Due Thursday, February 26 at 11:59 p.m.

The 9 hour grace period does apply to this problem set. Therefore, you can submit before Friday, Feb. 27 at 8:59 a.m. without any late penalty. After that point, there is late penalty (see syllabus).

You make work with others. If you work with others, please make sure you include whom you worked with. You are responsible for writing your own solutions and code.

You may use the help documentation in [R](#), and you may use the internet if you need it. If you use the internet (including gen AI tools), you must cite the websites you used by including the URL.

See the course's generative artificial intelligence policy in the syllabus for more details about what gen AI use is acceptable.

Instructions

For this problem set, please answer the questions in a Quarto document. Your work must be organized (using headers and with work for questions appearing in order) and you must include BOTH CODE AND OUTPUT.

On Canvas, you will upload 2 files: a PDF document with your answers and a txt file of your code.

Remember you can add this code chunk to your Quarto document to save your code as a txt file.

```
```${r}
#| include: false
courseTools::saveCode(last_name = '') # add your last name in the quotes
```
```

Feel free to email me with questions or issues with your code.

(The questions begin on the next page.)

Question 1 builds upon the activity we started in class on Feb. 16.

Question 1

This question uses the Insurance data set from the Feb. 16 activity. Use the train and validation data frames you created from that activity to complete the following questions.

Part (a).

Define a self-defined R function for R^2 . (Use the formulas for RSS and TSS from the slides.)

Part (b).

Define self-defined R functions for MSE, RMSE, and MAE. (Use the formulas from the slides.)

It is okay to define RMSE to be a function of MSE.

Part (c).

List 5 unique combinations of the available predictors that you will explore some multiple linear regression models for charges.

For example, in class we just used age as a predictor: charges ~ age. Another combination might be to use all the available predictors.

Part (d).

Fit a lm() on the train data frame for each of your 5 combinations. For each combination, report the R^2 and R^2_{adj} . The R^2 should be computed using your self defined function from part (a); however, the R^2_{adj} can come directly from the summary output.

Part (e).

Compute the MSE, RMSE, and MAE based on the validation data frame for each of the 5 combinations you chose. Report your answers in a table; i.e., something like this:

| Predictors | Validation Set MSE | Validation Set RMSE | Validation Set MAE |
|------------|--------------------|---------------------|--------------------|
| | | | |
| | | | |
| | | | |
| | | | |

Part (f).

Which model gives the best MSE based on the validation set?

Question 2

Suppose you are given the following `lm()` summary output.

```
Call:
lm(formula = Price ~ Miles + Model, data = Veh_train)

Residuals:
    Min       1Q   Median       3Q      Max
-8.215 -2.317 -0.647  1.929  8.480

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  18.69496    0.94390   19.806 < 2e-16 ***
Miles        -0.07807    0.01075   -7.262 3.87e-10 ***
ModelTacoma   9.45872    0.93403   10.127 2.02e-15 ***
ModelVolt    -11.02341    1.21108   -9.102 1.52e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.45 on 71 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared:  0.8313,    Adjusted R-squared:  0.8241
F-statistic: 116.6 on 3 and 71 DF,  p-value: < 2.2e-16
```

You are also given the validation set, which is available here:

https://raw.githubusercontent.com/maxwellhahn/maxwellhahn/refs/heads/main/Validation_set.csv

Based on the summary output and the validation set, compute the predictions for `Price` that the model would make for each row of the validation set.

Then, compute the RMSE, also using the ground truth value for `Price` contained in the csv.

Questions 3-5 build upon the activity we started in class on Feb. 9.

Question 3

The file NOVA.csv contains a sample of properties in the Northern Virginia region (Arlington County, Fairfax County, and so on) listed on Remax.com in April 2019. The data set contains the following variables:

- Bedrooms - number of bedrooms
- Bathrooms - number of bathrooms
- LivingSQFT - square footage of living area
- Age - age of the house in years
- DistToWH - distance from house to the White House in miles
- LotSize - lot size in acres
- ListingType - type of property (Condo/Townhome OR Single Family)
- Floors - the number of floors in the house
- CountySchool - the school district for the country
- Garage - the number of garage stalls
- Price - the price of the home in dollars

Part (a).

We are going to perform some data wrangling procedures on the dataset. Since the dataset is relatively small, we will keep a copy of the original dataset and make modifications to a copy named `NOVA2`. At this time, create a dataset named `NOVA2` that is a copy of `NOVA`. Use `NOVA2` in the remaining parts of Problem 3.

Part (b).

We can think of CountySchool as a proxy for the property's county. How many properties are included from each county?

Part (c).

Use the code below to generate a histogram of Price using a binwidth of 200000. The distribution of house prices is right skewed with several outliers. The heavy red line represents a price of three million dollars. Properties with a price over 3 million dollars are emphasized by the red circles. It looks like there are at least 5 properties with a price of more than \$3M.

(code on next page)

```
#Create histogram of Prices and highlight outliers
ggplot(data = NOVA2, mapping = aes(x = Price)) +
  geom_histogram(binwidth = 200000, color = "black", fill = "lightblue") +
  labs(x = "Price (in Dollars)",
       y = "Number of Properties") +
  #Removes scientific notation from axis labels
  scale_x_continuous(labels = scales::comma) +
  geom_vline(xintercept = 3000000, color = "red", linewidth = 2) +
  geom_point(data = NOVA2[NOVA2$Price > 3000000,], aes(x = Price, y = 0),
            color = "red", shape = 1, size = 10)
```

Part (d).

To investigate what makes these properties unusual, create and display a table showing the properties with a price of more than \$3M. (Do NOT overwrite NOVA2 with a sorted version of the dataset when doing this step.) What do you notice about these properties?

Part (e).

We are going to refine the scope of our analysis. We only want to model properties worth less than \$3M from Arlington and Fairfax counties. Since Loudoun and Prince William counties have very few observations (7 and 1, respectively) and we have not collected enough data from very expensive properties (only 5 properties over \$3M), we do not really have enough data to accurately model these properties. Write code that removes the outliers (prices over \$3M) AND any properties from Loudoun and Prince William counties from NOVA2. How many rows remain in NOVA2?

Part (f).

In modeling housing prices, it is common to do a log (natural) transformation of the Price. Create a new variable called logPrice in NOVA2 that represents the natural log of the house's price, create a histogram of logPrice using ggplot, modify the binwidth so that you are not using the default plot, and comment on the shape of the distribution.

Part (g).

Create indicator variables for ListingType and CountySchool. In particular, create a variable in NOVA2 called **Single** that takes a value of 1 if the property is a "Single Family" home and a value of 0 otherwise. Similarly, create a variable called **Arlington** that takes a value of 1 when the county school is "ARLINGTON COUNTY PUBLIC SCHOOLS" and 0 otherwise.

Note

NOTE: To verify that you have performed these steps correctly, double check that the mean of log price is 13.50511 and the mean of Arlington is 0.2637151 (which is the proportion of properties in NOVA2 from Arlington). If you are NOT getting those EXACT (not close) answers, it will affect future answers. You should fix this before moving forward.

Question 4

Part (a).

Suppose we want to build a simple linear regression model for predicting `logPrice` as a function of `Bathrooms` using the `NOVA2` dataset created through the various steps in Problem 3. We should first determine if a linear relationship is appropriate. Create a scatterplot showing the relationship between the variables and include a linear smoother. Do you think a linear relationship is appropriate?

Part (b).

Calculate the correlation between `logPrice` and `Bathrooms`.

Part (c).

Build the regression model, display a summary of the model, and write the estimated regression equation.

Note

If you are working in a Quarto/RMD document, you can try to type out the equation using LaTeX by including this in your document:

```
$$  
\hat{y}_i = 2 + 3x_{i,\text{baths}}  
$$
```

Part (d).

Report and interpret R^2 for this model.

Part (e).

What is the average amount by which the `logPrice` will deviate from the prediction made by the regression line?

Part (f).

(Challenge) Recreate the plot from Part a. Then, draw a red line segment on the plot that represents the residual for the observation with the largest number of bathrooms. NOTE: We have not covered a method for drawing a line segment on a plot. You will either have to build on your STAT 184 knowledge or rely on a web search for this.

Question 5

Part (a).

Using `NOVA2`, which you created in Problem 3, build a regression model for `logPrice` using `Bathrooms` and the `Arlington` indicator. Display a summary of the model, and write the estimated regression equation.

Part (b).

Using the estimated regression equation from Part a., predict the `logPrice` of a house with 2 bathrooms in Fairfax County. NOTE: I want you to show your work and do this calculation manually rather than writing R code to predict this value. (I am trying to give you practice for something I could ask on a pencil/paper exam.)

Part (c).

Explain the relationship between the number of bathrooms and `logPrice` after accounting for location (`Arlington` indicator).

Part (d).

Interpret the coefficient (or slope) with `Arlington` in the context of the problem.

Part (e).

Report a 90% confidence interval the coefficient (or slope) for `Bathrooms`.

Part (f).

What proportion of the total variation in `logPrice` is explained by the model using `Bathrooms` and `Arlington`?

Part (g).

Report R^2_{adj} for this model.